

Muhua (黄牧华) Huang

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Computational social scientist and Stanford Organizational Behavior PhD researcher studying how AI systems represent human traits and reshape collective behavior. I build large-scale LLM-agent experiments and psychometric methods to investigate human-AI collaboration, culture, values, and emergent social dynamics.

EDUCATION

Stanford Graduate School of Business, PhD in Organizational Behavior	2025 – 2030
University of Chicago, MA in Computational Social Science, GPA: 3.9/4.0	2023 – 2025
University of British Columbia, BA in Computer Science & Psychology (Honours), GPA: 4.2/4.3	2018 – 2023

Additional training: International AI Evaluation Programme Fellow; SICSS-Beijing (Outstanding Project, scholarship); Google Research Mentorship Program Scholar.

RESEARCH EXPERIENCE

PhD Student Researcher | Stanford Graduate School of Business 2025 – Present

- Built a large-scale multi-agent coordination experiment with approximately 375,000 GPT-4o trials across networks of 2–50 agents, replicating a human category-formation paradigm.
- Identified systematic human-AI differences in category boundaries and showed that constraining agents to human labels reduced performance by 12.7%.

Master's Researcher | University of Chicago 2023 – 2025

- Developed a psychometrically validated framework for assigning Big Five personalities to LLM agents across GPT-3.5, GPT-4, and Claude 3.
- Collected data from 350 participants and showed that assigned personalities predict agents' moral reasoning, risk-taking, and cooperation; published in *Personality Science*.

Research Assistant | University of British Columbia 2020 – 2023

- Used SEM and simulation to evaluate alternative personality scale formats and CFA fit indices; contributed to publications in *Journal of Personality Assessment* and *Psychological Methods*.
- Applied ML/NLP to 10,000+ youth essays, supervised 20 research assistants, and studied personalized AI explanations using eye-tracking; contributed to *Journal of Social Issues* and an IJCAI XAI workshop paper.

INDUSTRY EXPERIENCE

Student Researcher | Google DeepMind Jun – Sep 2026

Research Intern | Microsoft Research Asia Jul – Oct 2024

SELECTED PUBLICATIONS & RESEARCH CONTRIBUTIONS

- Huang, M.**, et al. (2026). Designing AI-Agents with Personalities: A Psychometric Approach. *Personality Science*.
- Huang, M.**, & Evans, J. (2026). Institutions as cached computation for resource-rational negotiation. *Behavioral and Brain Sciences*, 49, e144.
- Bai, Y., Duan, S., **Huang, M.**, et al. (2026). IROTE: Human-like Traits Elicitation of LLMs via In-Context Self-Reflective Optimization. *AAAI*.
- Savalei, V., & **Huang, M.** (2025). Fit Indices Are Insensitive to Multiple Minor Violations of Perfect Simple Structure in Confirmatory Factor Analysis. *Psychological Methods*.
- Stanford HAI & Google DeepMind** (2026). *Organizing Intelligence*. Contributing researcher and writer; acknowledged for conceptualization and research.

LEADERSHIP & SERVICE

Computational Social Science Workshop Coordinator | University of Chicago 2024 – 2025
Organized the weekly MACSS workshop featuring speakers on advanced computational methods and interdisciplinary research.

Teaching Assistant | University of British Columbia 2021 – 2022
Taught Human Computer Interaction Methods and Systematic Program Design; rated 4.7/5 and received three return offers.

Crisis Respondent | Kids Help Phone; 230+ hours and 550+ people supported 2019 – 2020

Peer Reviewer | *Personality Science*; *Behaviour & Information Technology*

Methods: LLM agents/GABM, psychometrics/SEM, social network analysis, ML/NLP, behavioral experiments, RAG, Python, R, SQL.

Recognition: Stanford GSB & EDGE Fellowships; UChicago Thesis Award; MSRA Outstanding Intern; OpenAI Researcher Access.